GREATER NOIDA

Department of Computer Science & Engineering

AICTE • EDUSKILLS • NATIONAL INTERNSHIP PORTAL

INTERNSHIP REPORT

IoT Cloud Engineer Virtual Internship

An 8-week virtual internship on cloud-enabled IoT engineering, with curriculum delivered through AWS Skill Builder under the AICTE–EduSkills National Internship Programme.

STUDENT NAME

Yash Tyagi

ENROLLMENT NUMBER

2410030590

PROGRAM & SEMESTER

B.Tech CSE — 5th Semester

June – August 2026

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What This Report Covers

PROGRAMME OVERVIEW

About the Internship

The IoT Cloud Engineer Virtual Internship is a National Internship Portal initiative run by AICTE in partnership with EduSkills, with technical curriculum delivered through AWS Skill Builder. The programme introduces students to cloud-enabled Internet of Things (IoT) engineering — connecting devices, moving their data into the cloud, and building secure, scalable services on top of it. It combines structured e-learning, hands-on labs, and a guided capstone project completed under faculty supervision.

It is designed to help engineering students bridge classroom concepts with the practical, industry-aligned skills expected of a cloud-facing IoT engineer.

8 wks — Programme Duration

100% — Virtual / Remote Mode

1 — Guided Capstone Project

Grade O — Outstanding Performance

WHO MADE IT POSSIBLE

Programme Partners

AICTE

All India Council for Technical Education administers the programme nationally through the National Internship Portal, and co-issues the final internship certificate.

EduSkills Foundation

EduSkills coordinates enrollment, mentoring sessions, faculty guidance, and project evaluation across participating institutions.

AWS Skill Builder

AWS Skill Builder supplies the technical curriculum — structured courses, hands-on labs, and assessments covering cloud and IoT concepts.

Together, these partners connect national policy, mentorship, and industry-grade cloud training into a single guided internship experience.

HOW IT UNFOLDED

Internship Timeline

1. **Registration**
   Applied via the AICTE National Internship Portal and the EduSkills portal.

2. **Enrollment**
   Shortlisted and enrolled on the AWS Skill Builder learning platform.

3. **Course Modules**
   Completed structured cloud & IoT modules with online assessments.

4. **Mentoring Sessions**
   Attended guided sessions with industry mentors and faculty.

5. **Capstone Project**
   Built and submitted a guided IoT-cloud project.

6. **Certification**
   Awarded the AICTE–EduSkills Internship Certificate, Grade O.

CURRICULUM

Modules Covered

- Cloud Computing Fundamentals
- Introduction to IoT Systems
- AWS IoT Core & Device Management
- Data Ingestion & Analytics
- Cloud Storage & Databases
- Networking & Cloud Security
- Architecting IoT-Cloud Solutions
- Capstone Project & Evaluation

TOOLS & PLATFORM

Core Technologies & AWS Services Explored

AWS IoT Core

Connecting and managing simulated IoT devices securely.

Amazon EC2

Provisioning virtual compute instances for backend services.

Amazon S3

Storing device data and static application assets.

AWS Lambda

Running serverless functions to process incoming data.

Amazon DynamoDB

Persisting device telemetry in a managed NoSQL store.

IAM & CloudWatch

Managing access control and monitoring system health.

APPLIED LEARNING

Hands-on Project Work

Alongside the course modules, the internship required a guided capstone project connecting a simulated IoT sensor pipeline to cloud services — completed under faculty supervision and reviewed during mentoring sessions with industry experts.

PROJECT SNAPSHOT

Cloud-Connected Environmental Monitoring System

A simulated sensor pipeline streams telemetry to AWS IoT Core, which routes data through a serverless function into a managed database, with a dashboard for monitoring and basic alerting.

KEY ACTIVITIES

- Simulated IoT device data streams
- Configured cloud ingestion & routing
- Built serverless processing logic
- Stored & queried telemetry data
- Applied access-control best practices
- Presented outcomes to mentors

OUTCOMES

Skills Gained

Technical Skills

- Core cloud computing concepts & AWS service fundamentals
- Connecting and managing IoT devices in the cloud
- Serverless data processing with AWS Lambda
- Structured & NoSQL data storage on AWS
- Cloud networking, IAM, and security best practices
- Monitoring systems with CloudWatch

Professional Skills

- Self-paced learning and time management
- Following documentation & technical guides independently
- Working with faculty and industry mentors
- Presenting technical work clearly
- Structuring and documenting a project report
- Problem-solving in unfamiliar tools and platforms

RECOGNITION

Certification & Outcomes

Course Completion Certificate

Issued on completing all AWS Skill Builder modules & assessments.

Digital Badge

A shareable credential recognizing the completed coursework.

AICTE–EduSkills Internship Certificate

The final internship certificate, jointly issued by AICTE and EduSkills.

Grade: O (Outstanding)

Awarded for overall performance across coursework and project.

Certificate ID: 4e85bfe5c9b9ac091025 • Student ID: STU6997df0952b451771560713

REFLECTION

Key Learnings

Cloud & IoT are inseparable

Real IoT systems depend on cloud services for storage, processing, and scale — the two can't be learned in isolation.

Security is designed in, not added later

Access control and monitoring shaped every stage of the project, not just the final step.

Documentation accelerates learning

AWS Skill Builder's structured guides made unfamiliar services approachable within an 8-week window.

Mentorship sharpens direction

Feedback from industry mentors and faculty was decisive in refining the capstone project.

CONCLUSION

A Strong Foundation in Cloud-Enabled IoT Engineering

The IoT Cloud Engineer Virtual Internship gave a structured, practical introduction to how connected devices and cloud infrastructure work together — from device onboarding to data processing, storage, and security. Completing it under the AICTE–EduSkills programme, with curriculum from AWS Skill Builder, has built a solid base for further study and future roles in cloud and IoT engineering.

CERTIFICATION

Certificate of Completion

Yash Tyagi successfully completed the 8-week IoT Cloud Engineer Virtual Internship with an Outstanding (Grade O) performance.

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THANK YOU

Yash Tyagi

IoT Cloud Engineer Virtual Internship — an 8-week programme completed at IILM University, Greater Noida.

Roll No. 2410030590 • B.Tech CSE, 5th Semester